Austin Amakye Ansah

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Summary

Reverse-engineered proprietary file formats to build Floss, a cross-platform painting app in C#. Solved real-time telemetry bottlenecks in CCrumb, an offline ESP32 tracing device, with a custom serial protocol handling millions of packets per trace. Led 700+ sessions teaching mobile app development from architecture to shipped product, while shipping client projects as an instructor.

Education

University of Texas at Arlington — Arlington, TX

B.S. Computer Science — *Aug. 2022 – May 2026*

Experience

Coding Instructor & Full Stack Software Engineer — May 2023 – Present

Coding Minds Academy

- Successfully led and coordinated 700+ student app and software development class sessions
- Designed and implemented hardware-software interfaces for IoT devices, utilizing C/Rust for performance-critical firmware and Dart/Flutter for client-facing applications.
- Co-authored 12+ student publications to CS & IT and various conferences.
- Developed full-stack Coding Minds client projects and contributed to internal brand website and course development

Language Model Trainer — May 2024 - Present

DataAnnotation

- Evaluated LLM outputs for correctness and instruction-following across data analysis and algorithm generation tasks; manually edited responses to produce high-quality RLHF training data for models widely used today, like Claude and Codex.
- Evaluated coding models on real use-cases and personal projects to create conversation pairs for fine-tuning
- Created edited photo dataset pairs for model image-editing capabilities

Projects

CCrumb · *github.com/breadcompass/ccrumb2* — C

- Low-power embedded device for retracing steps in the wilderness built on the ESP32 platform with FreeRTOS
- Developed low-level peripheral hardware drivers and desktop telemetry tooling to establish the foundational data pipeline.
- Authored a modified XMODEM serial data transfer protocol over USB to ensure resilient packet transmission between embedded hardware and desktop host applications.
- Integrated the Rerun visual logging framework to stream and visualize real-time IMU sensor fusion and GPS telemetry, accelerating hardware debugging cycles by over 10x
- Designed a compact binary format capable of serializing millions of packets of data per trace.

Floss · *gitlab.com/Skynse/floss* — C#

- Cross-platform image editing and digital painting software
- Serves as a Krita/Clip Studio Paint alternative
- Reverse-engineered the Photoshop brush, and Clip Studio File format to bring their shared usefulness to one platform.
- Incorporated the Avalonia UI framework to establish a modern, intuitive UI

QPitch · *github.com/Skynse/QPitch* — C++

- Free Cross-platform Autotune/pitch correction plugin that competes with popular plugins like Antares Autotune and Waves Autotune.
- Built on the JUCE framework

Publications

CS & IT Conference Proceedings

- Derek Li and **Austin Amakye Ansah**. *HoopLab: A Cross-Platform YOLO-Powered Mobile Application for Real-Time Basketball Shot Detection and Trajectory Analysis*. Vol. 15, No. 24.
- Nina Zhang and **Austin Amakye Ansah**. *Smart Vehicle Recognition and School Pickup Efficiency Enhancing System using Artificial Intelligence and IoT System*. Vol. 15, No. 4.

Tooling

Languages: Dart, Rust, Python, C, C++, Java, TypeScript, C#

Frameworks & Tools: Flutter, React, Next.js, PostgreSQL, Git, Docker/Podman

Machine Learning: TensorFlow, PyTorch

OS: Linux